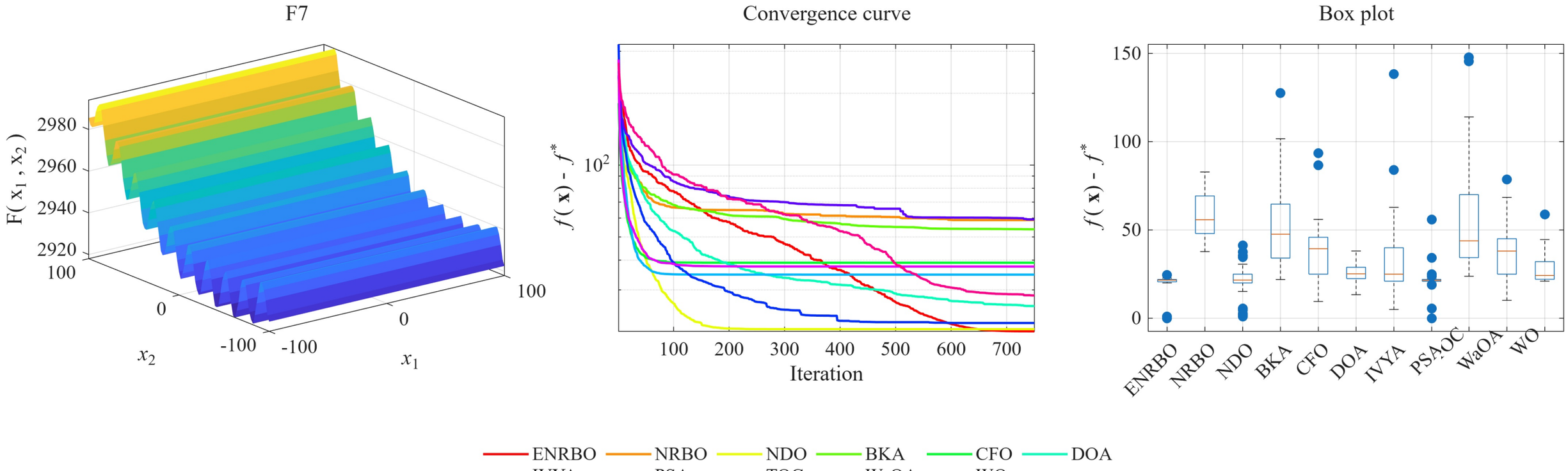

F7
F(x1 , x2)
2980
2960
2940
2920
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
10^2
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
150
100
50
0
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
OC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO